

Academic Tasks CSE326 (26271)

Student Name: _____

Reg. Number: _____

Section: _____

Max Marks: 30

Question 1 — DevPulse — Developer Curriculum & Syllabus Hub

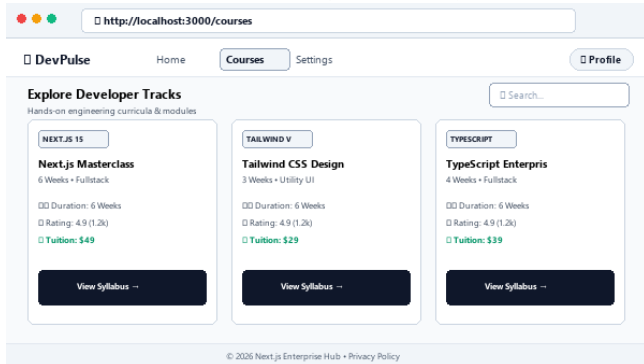
Scenario: You are developing the curriculum portal for DevPulse. Users browse technical learning tracks on a catalog page, navigate to individual course syllabi via URL slugs, interactively bookmark tracks on the client, and see graceful shimmer loading skeletons and custom 404 not-found fallbacks. Data is fetched server-side from a mock curriculum API.

Target Route Architecture & Expected Functionality:

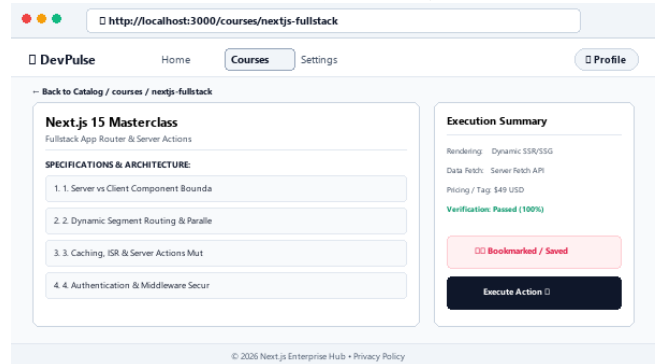
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/courses	Server-Side Rendered (SSR) catalog grid displaying tech tracks fetched from '/api/courses' with difficulty badges and live search filter.
/courses/nextjs-fullstack	Resolves the dynamic course track to render the complete module syllabus, tuition details, and an interactive client bookmark button.
/courses/invalid-course-slug	Gracefully displays a custom 404 Not Found fallback UI when a non-existent track is requested.

Given Visual Outputs (Reproduce Exactly as Shown Below):

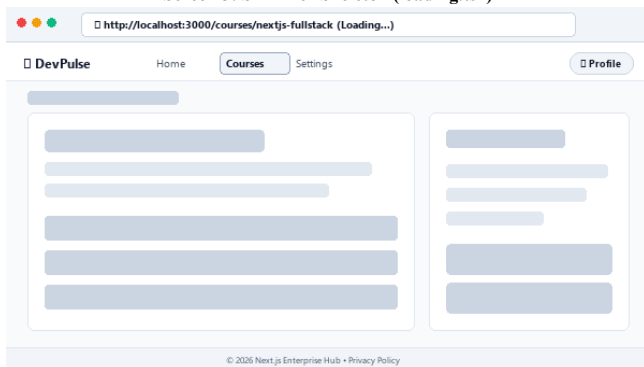
Screen 1: Primary View (/courses)



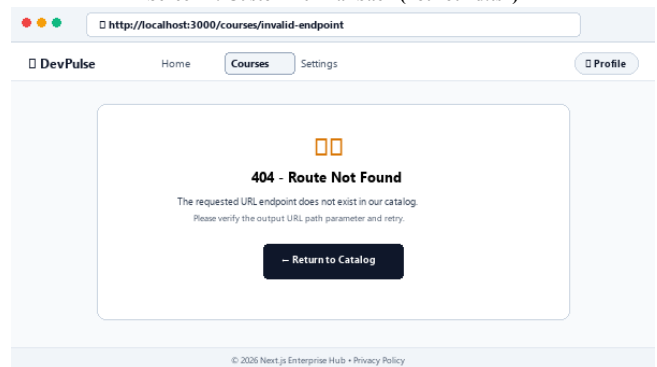
Screen 2: Detail View (/courses/nextjs-fullstack)



Screen 3: Shimmer Skeleton (loading.tsx)



Screen 4: Custom 404 Fallback (not-found.tsx)



Academic Tasks CSE326 (26271)

Question 2 — OmniDoc — Medical Specialty Directory & Doctor Profiles

Scenario: Build a high-performance hospital directory where medical departments and doctor profiles are pre-rendered at build time using Static Site Generation (SSG with `generateStaticParams`) from `https://dummyjson.com/users`, supporting instant navigation and interactive consultation slot selection on the client.

Target Route Architecture & Expected Functionality:

Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/specialties	Statically generated (SSG) department directory grid displaying certified medical fields (Cardiology, Neurology, Pediatrics).
/specialties/cardiology/dr-sarah-smith	Pre-rendered clinician profile displaying medical credentials, certifications, and interactive appointment slot picker.
/specialties/cardiology/unregistered-doctor	Triggers custom 404 fallback for unlisted doctor entries.

Given Visual Outputs (Reproduce Exactly as Shown Below):

Screen 1: Primary View (/specialties)

http://localhost:3000/specialties

OmniDoc

HomeDoctorsSettings

Profile

Medical Specialty Directory (SSG)

Certified clinicians & specialist departments

Filter records...

ID / CODE	RECORD / DESCRIPTION	STATUS	ACTION
DOC-101	Dr. Sarah Smith, MD (Cardiology)	Active	Book Slot --
DOC-102	Dr. James Wilson, MD (Neurology)	Active	Book Slot --
DOC-103	Dr. Elena Rostova, MD (Pediatric)	Active	Book Slot --
DOC-104	Dr. Michael Chang, MD (Oncology)	Active	Book Slot --

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Screen 2: Detail View (/specialties/cardiology/dr-sarah-smith)

http://localhost:3000/specialties/cardiology/dr-sarah-smith

OmniDoc

HomeDoctorsSettings

Profile

Specialties / cardiology / dr-sarah-smith

Dr. Sarah Smith, MD

Chief of Cardiology • 15 Yrs Experience

SPECIFICATIONS & ARCHITECTURE:

- Pre-rendered via generateStaticPar
- Clinic: Heart Care Center
- Certifications: Board Certified Ca
- Next Available: Today 2:30 PM

Execution Summary

Rendering: Dynamic SSR/SSG

Data Fetch: Server Fetch API

Pricing / Tag: \$120 Consultat

Verification: Passed (100%)

Bookmarked / Saved

Execute Action

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Screen 3: Shimmer Skeleton (loading.tsx)

http://localhost:3000/specialties/cardiology/dr-sarah-smith (Loading...)

OmniDoc

HomeDoctorsSettings

Profile

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Screen 4: Custom 404 Fallback (not-found.tsx)

http://localhost:3000/specialties/invalid-endpoint

OmniDoc

HomeDoctorsSettings

Profile

404 - Route Not Found

The requested URL endpoint does not exist in our catalog. Please verify the output URL path parameter and retry.

Return to Catalog

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Academic Tasks CSE326 (26271)

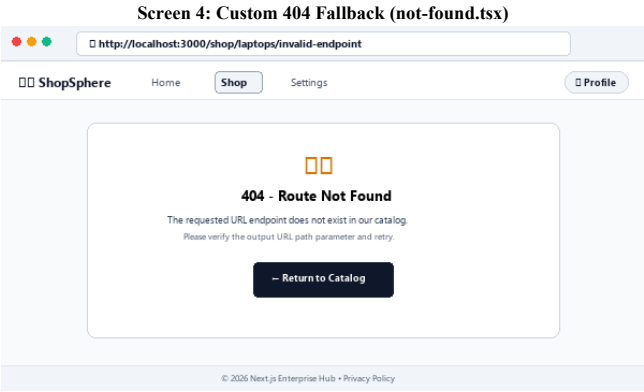
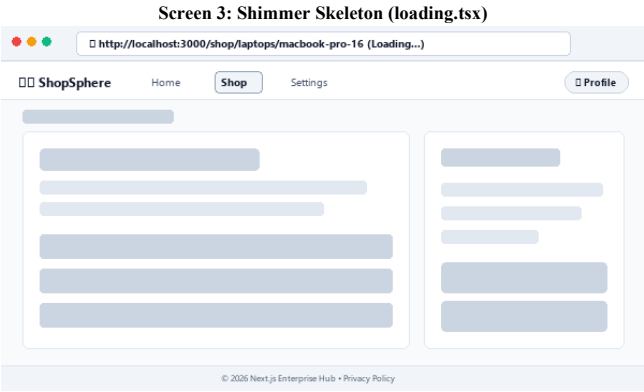
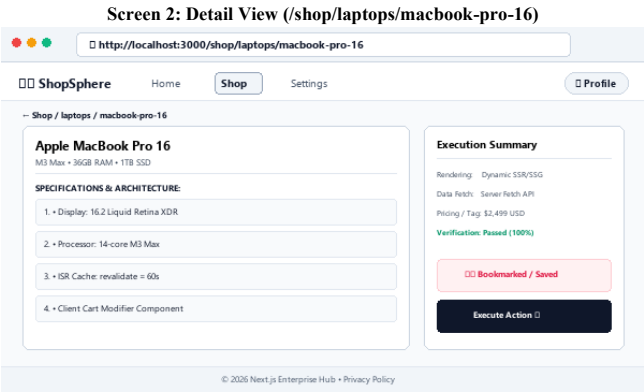
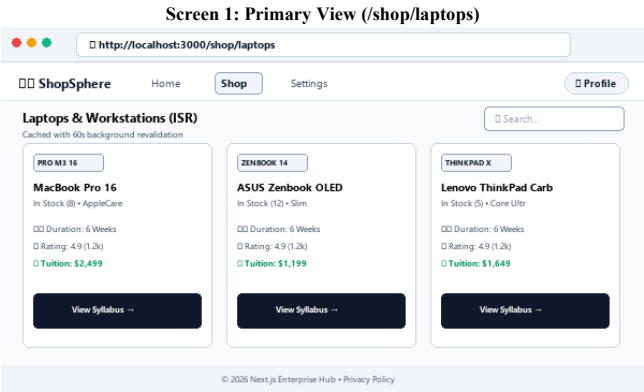
Question 3 — ShopSphere — E-Commerce Catalog with Background Revalidation

Scenario: Develop an e-commerce catalog leveraging Incremental Static Regeneration (ISR with `revalidate: 60`) to cache product listings from `https://dummyjson.com/products`, automatically revalidating fresh inventory in the background while users adjust cart quantities on the client.

Target Route Architecture & Expected Functionality:

Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/shop/laptops	ISR-cached category product catalog with stock badges, live search, and price sorting filters.
/shop/laptops/macbook-pro-16	Product specification view rendering high-resolution details and interactive 'Add to Cart' quantity controller.
/shop/laptops/discontinued-model	Gracefully displays custom 404 fallback for discontinued product items.

Given Visual Outputs (Reproduce Exactly as Shown Below):



Academic Tasks CSE326 (26271)

Question 4 — DocuMatrix — Hierarchical Technical Documentation Explorer

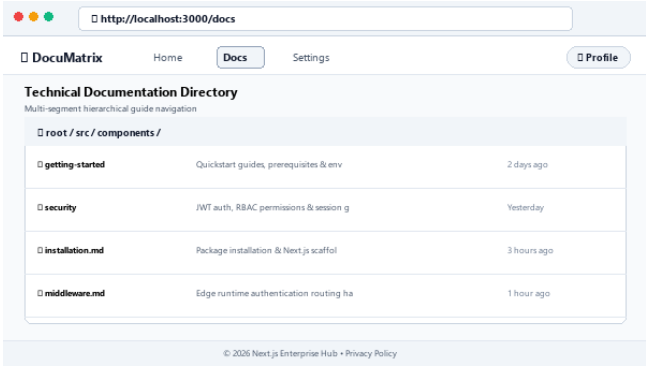
Scenario: Create a multi-level developer documentation explorer supporting arbitrarily deep guide hierarchies, dynamic breadcrumb trails, and an interactive copy-snippet action with layout continuity across nested folders.

Target Route Architecture & Expected Functionality:

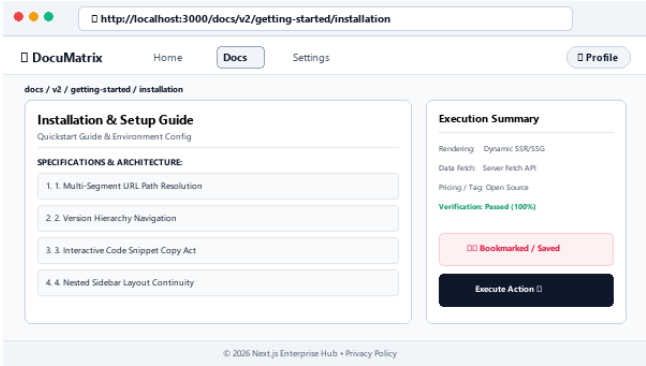
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/docs	Landing page highlighting quickstart guides and API reference categories.
/docs/v2/getting-started/installation	Resolves multi-segment guide paths to render markdown-formatted technical instructions.
/docs/v2/security/jwt-tokens	Direct deep link to authentication verification architecture with interactive code snippet copy.

Given Visual Outputs (Reproduce Exactly as Shown Below):

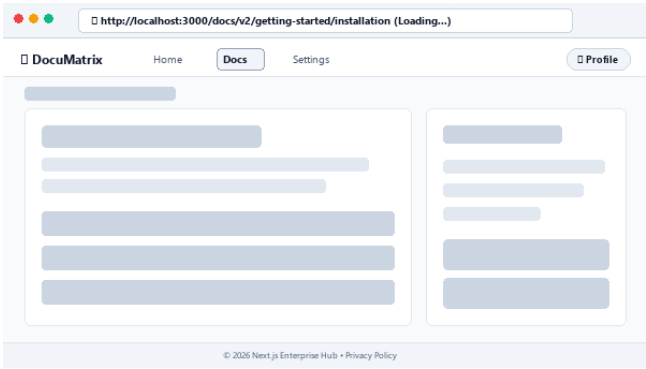
Screen 1: Primary View (/docs)



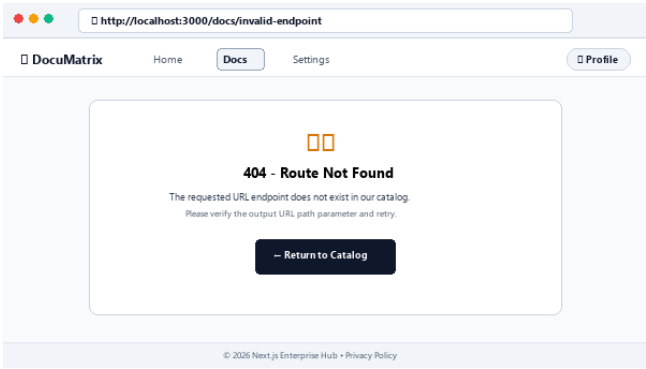
Screen 2: Detail View (/docs/v2/getting-started/installation)



Screen 3: Shimmer Skeleton (loading.tsx)



Screen 4: Custom 404 Fallback (not-found.tsx)



Academic Tasks CSE326 (26271)

Question 5 — TravelZen — Flight Matrix with Hybrid Data Fetching

Scenario: Build a travel portal demonstrating hybrid data fetching: flight schedules are fetched server-side from a mock airline API with URL query synchronization (`?from=SFO&to=JFK`), while real-time seat availability is fetched client-side upon user selection.

Target Route Architecture & Expected Functionality:

Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/search?from=SFO&to=JFK&sort=price	Server-Side Data Fetching filtering available flights based on URL search query parameters.
/flights/DL-402/seats	Interactive aircraft cabin seat map selection with real-time seat lock state.
/flights/CANCELLED-999	Gracefully displays custom 404 for cancelled or invalid flight numbers.

Given Visual Outputs (Reproduce Exactly as Shown Below):

Screen 1: Primary View (/search?from=SFO&to=JFK)

The screenshot shows a web application interface for 'TravelZen'. The top navigation bar includes 'Home', 'Flights' (active), 'Settings', and 'Profile'. The main content area displays 'SFO → JFK Flight Schedule Matrix' with a sub-header 'Direct nonstop flights • Sorted by price'. Below this is a table with columns: ID / CODE, RECORD / DESCRIPTION, STATUS, and ACTION. The table lists four flights: DL-402 (Delta Air Lines, 08:30 AM, 5h 3), UA-118 (United Airlines, 10:15 AM, 5h 4), AA-094 (American Airlines, 01:00 PM, 5h), and B6-520 (JetBlue Airways, 03:45 PM, 5h 4). Each flight has a status of 'On Time' and a 'Select' button. A footer note reads '© 2026 Next.js Enterprise Hub • Privacy Policy'.

Screen 2: Detail View (/flights/DL-402/seats)

The screenshot shows the 'Flight DL-402 Seat Selection' page. The top navigation bar is the same as Screen 1. The main content area has a sub-header 'Search / SFO-JFK / DL-402 / seats'. Below this is a section titled 'Flight DL-402 Seat Selection' with a sub-header 'Boeing 777-300ER • Main Cabin'. The 'SPECIFICATIONS & ARCHITECTURE' section lists four items: 1. Server-Side Query Fetching (?from=), 2. Client-Side Seat Map SWR Fetch, 3. Departure: 08:30 AM (SFO Gate B 14), and 4. Selected Seat: 14A (Window). To the right is an 'Execution Summary' box showing 'Rendering: Dynamic SSR/SSG', 'Data Fetch: Server Fetch API', 'Pricing / Tag: \$349 USD', and 'Verification: Passed (100%)'. Below this is a 'Bookmarked / Saved' button and an 'Execute Action' button. A footer note reads '© 2026 Next.js Enterprise Hub • Privacy Policy'.

Screen 3: Shimmer Skeleton (loading.tsx)

The screenshot shows a 'Shimmer Skeleton' loading state for the flight details page. The top navigation bar is the same as Screen 1. The main content area is a placeholder with several horizontal bars of varying lengths, representing the layout of the flight details. A footer note reads '© 2026 Next.js Enterprise Hub • Privacy Policy'.

Screen 4: Custom 404 Fallback (not-found.tsx)

The screenshot shows a '404 - Route Not Found' page. The top navigation bar is the same as Screen 1. The main content area has a sub-header 'http://localhost:3000/search?from=SFO&to=JFK/invalid-endpoint'. Below this is a section titled '404 - Route Not Found' with a sub-header 'The requested URL endpoint does not exist in our catalog. Please verify the output URL path parameter and retry.' Below this is a 'Return to Catalog' button. A footer note reads '© 2026 Next.js Enterprise Hub • Privacy Policy'.

Academic Tasks CSE326 (26271)

Question 6 — TaskFlow — Agile Kanban Board with Server Actions & Optimistic UI

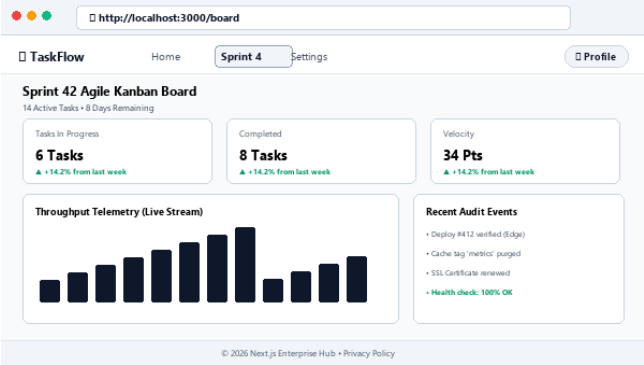
Scenario: Construct an agile sprint board where tasks are rendered across status columns (To Do, In Progress, Done), and moving a task triggers a Server Action with instant optimistic UI update (`useOptimistic`) and automatic path revalidation.

Target Route Architecture & Expected Functionality:

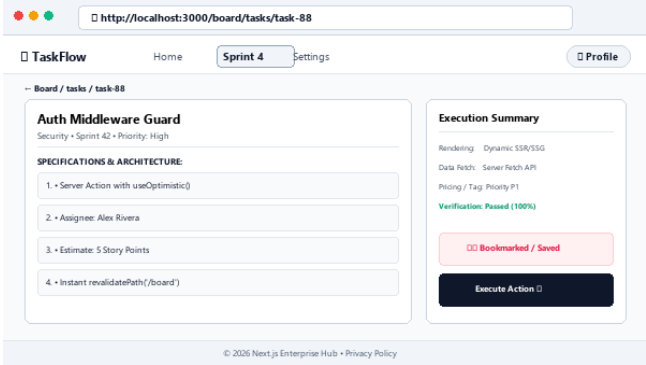
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/board	Sprint kanban board organizing tasks into status columns with live column counters.
/board/tasks/task-88	Task inspection view displaying assignee details, estimate story points, and priority badges.
/api/tasks/move	Server Action handling task status updates with optimistic UI and path revalidation.

Given Visual Outputs (Reproduce Exactly as Shown Below):

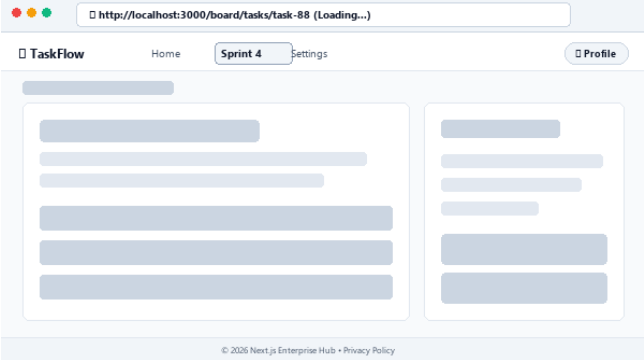
Screen 1: Primary View (/board)



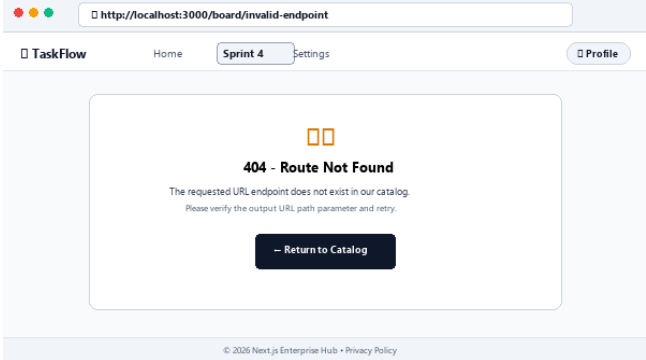
Screen 2: Detail View (/board/tasks/task-88)



Screen 3: Shimmer Skeleton (loading.tsx)



Screen 4: Custom 404 Fallback (not-found.tsx)



Academic Tasks CSE326 (26271)

Question 7 — FinTrack — Live Financial Markets with Dynamic SSR (No-Store)

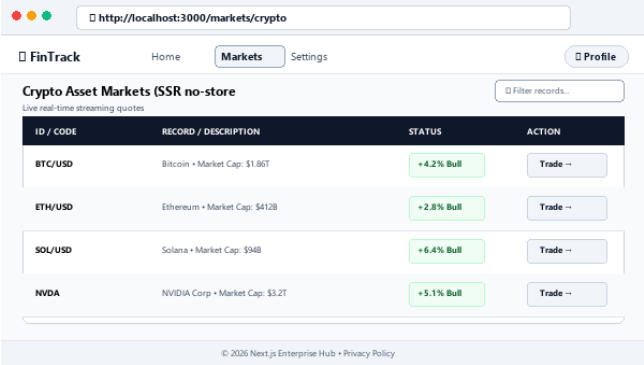
Scenario: Develop a financial market tracker enforcing pure Server-Side Rendering (SSR) per request using un-cached live market data (`cache: 'no-store'`) to deliver real-time asset quotes with search filtering and client watchlist actions.

Target Route Architecture & Expected Functionality:

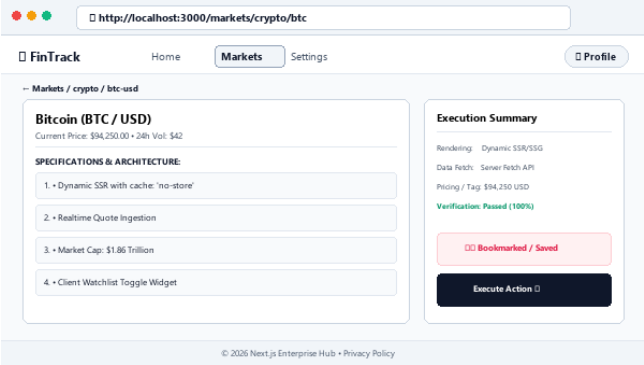
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/markets/crypto	Dynamic SSR market quote table with un-cached real-time pricing and 24h change indicators.
/markets/crypto/btc	Asset detail page rendering live order depth, market cap, and interactive watchlist toggle.
/markets/crypto/UNKNOWN-COIN	Custom 404 Not Found error screen for unlisted ticker symbols.

Given Visual Outputs (Reproduce Exactly as Shown Below):

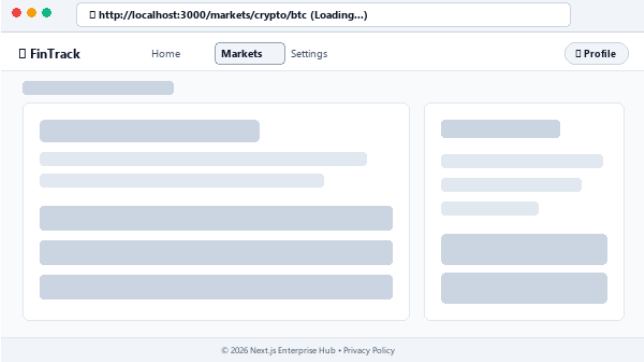
Screen 1: Primary View (/markets/crypto)



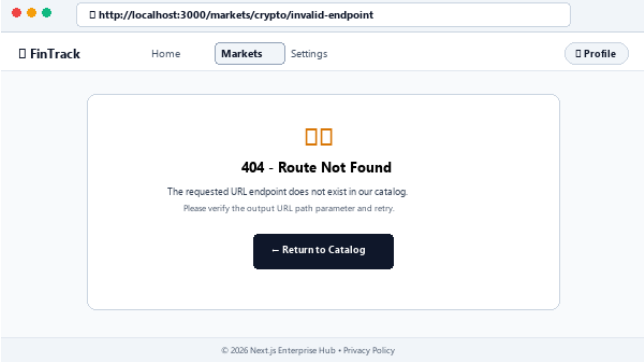
Screen 2: Detail View (/markets/crypto/btc)



Screen 3: Shimmer Skeleton (loading.tsx)



Screen 4: Custom 404 Fallback (not-found.tsx)



Academic Tasks CSE326 (26271)

Question 8 — DevSnippet — Code Playground with Parallel Version Comparison

Scenario: Build a code snippet hub where syntax-highlighted files are browsed by language tags, the code inspection page renders an independent parallel version comparison panel concurrently with the primary code view, and users execute fork actions.

Target Route Architecture & Expected Functionality:

Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/snippets/typescript	Language snippet catalog (TypeScript, Rust, Python) fetched server-side with fork stats.
/snippets/typescript/use-debounce	Primary code inspector view displaying syntax-highlighted code blocks and author profile.
/snippets/typescript/use-debounce/diff	Version diff comparison panel rendered concurrently with the primary editor.

Given Visual Outputs (Reproduce Exactly as Shown Below):

Screen 1: Primary View (/snippets/typescript)

http://localhost:3000/snippets/typescript

DevSnippetHomeSnippetsSettingsProfile

TypeScript Code Playground

Curated community utilities & algorithms

root / src / components /

useDebounce.ts

React 19 Custom Debounce Hook utility

1.4k Forks

zodParser.ts

Schema validation & error boundary map

920 Forks

serverAction.ts

Next.js Server Action form wrapper

640 Forks

queueWorker.ts

Edge Redis async queue worker utility

510 Forks

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Screen 2: Detail View (/snippets/typescript/use-debounce)

http://localhost:3000/snippets/typescript/use-debounce

DevSnippetHomeSnippetsSettingsProfile

Snippets / typescript / use-debounce

useDebounce Hook (TS 5.5)

TypeScript • Zero Dependencies

SPECIFICATIONS & ARCHITECTURE:

1. • Parallel View: Version Diff Inspec

2. • Target: React 18 / 19 Server Compo

3. • License: MIT Open Source

4. • Client Fork Counter Action

Execution Summary

Rendering: Dynamic SSR/SSG

Data Fetch: Server Fetch API

Pricing / Tag: MIT license

Verification: Passed (100%)

Bookmarked / Saved

Execute Action

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Screen 3: Shimmer Skeleton (loading.tsx)

http://localhost:3000/snippets/typescript/use-debounce (Loading...)

DevSnippetHomeSnippetsSettingsProfile

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Screen 4: Custom 404 Fallback (not-found.tsx)

http://localhost:3000/snippets/typescript/invalid-endpoint

DevSnippetHomeSnippetsSettingsProfile

404 - Route Not Found

The requested URL endpoint does not exist in our catalog.
Please verify the output URL path parameter and retry.

Return to Catalog

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Academic Tasks CSE326 (26271)

Question 9 — PulsePoll — Real-Time Community Polling with On-Demand Revalidation

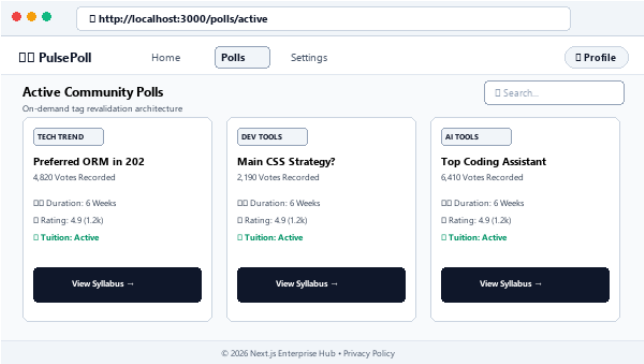
Scenario: Construct an opinion polling app where cached active polls are served with ISR tags, and voting triggers a Server Action that invokes tag-based on-demand revalidation to instantly broadcast updated percentage bars.

Target Route Architecture & Expected Functionality:

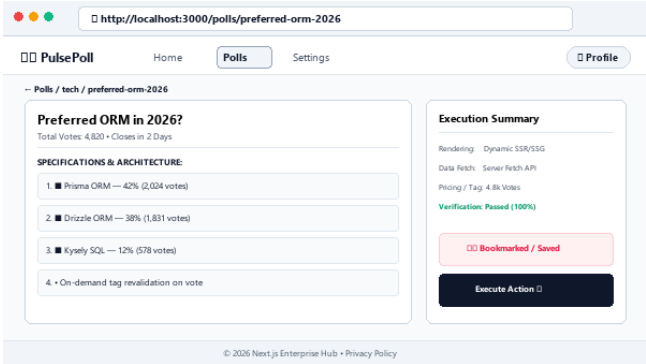
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/polls/active	ISR-cached polling feed displaying question topics and total votes recorded.
/polls/preferred-orm-2026	Interactive poll view where users cast ballots and view live percentage distribution bars.
/api/polls/vote	Server Action processing ballot submission with on-demand tag revalidation.

Given Visual Outputs (Reproduce Exactly as Shown Below):

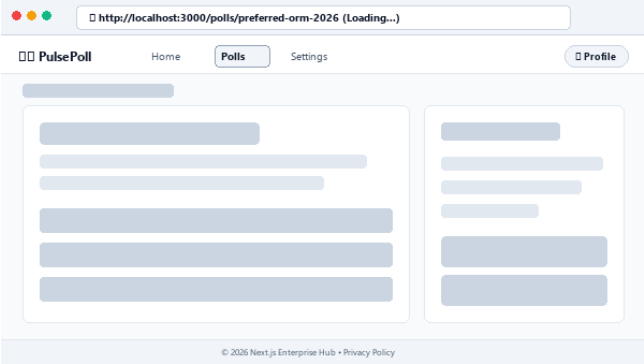
Screen 1: Primary View (/polls/active)



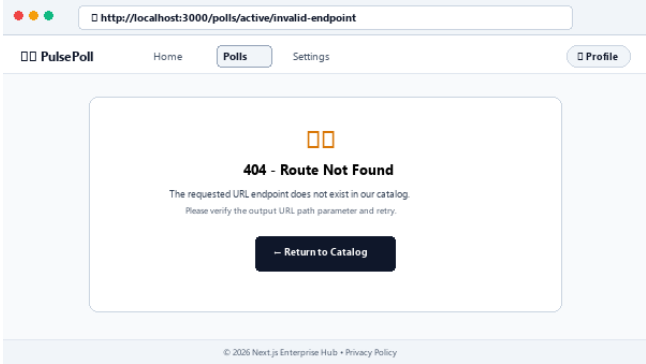
Screen 2: Detail View (/polls/preferred-orm-2026)



Screen 3: Shimmer Skeleton (loading.tsx)



Screen 4: Custom 404 Fallback (not-found.tsx)



Academic Tasks CSE326 (26271)

Question 10 — AudioSphere — Podcast Streamer with Persistent Layout Player

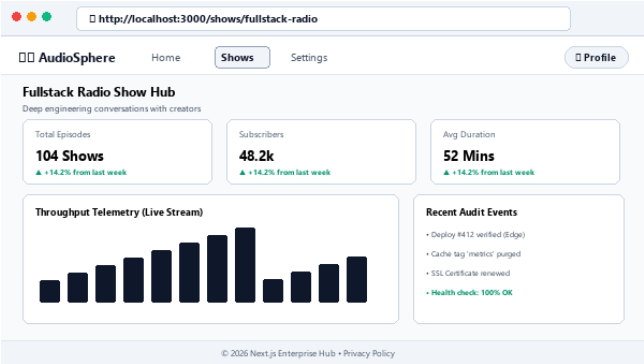
Scenario: Build a podcast listening web app where episode metadata is fetched server-side, while a floating audio player persists in the root layout maintaining uninterrupted audio playback across all route transitions.

Target Route Architecture & Expected Functionality:

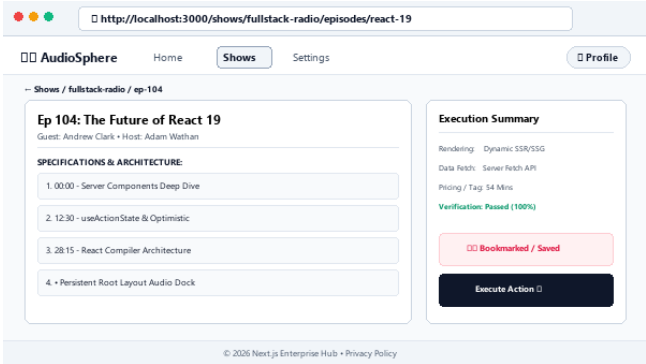
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/shows/fullstack-radio	Podcast show catalog with episode list, guest host badges, and play triggers.
/shows/fullstack-radio/episodes/react-19	Detailed episode notes view with topic timestamps and guest biography.
/shows/fullstack-radio/episodes/missing-audio	Custom error and not-found fallback for unavailable episode recordings.

Given Visual Outputs (Reproduce Exactly as Shown Below):

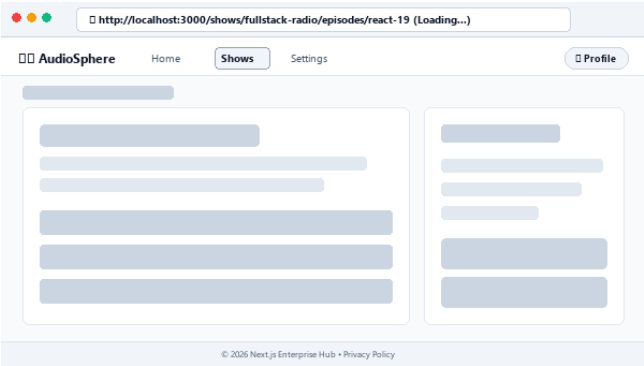
Screen 1: Primary View (/shows/fullstack-radio)



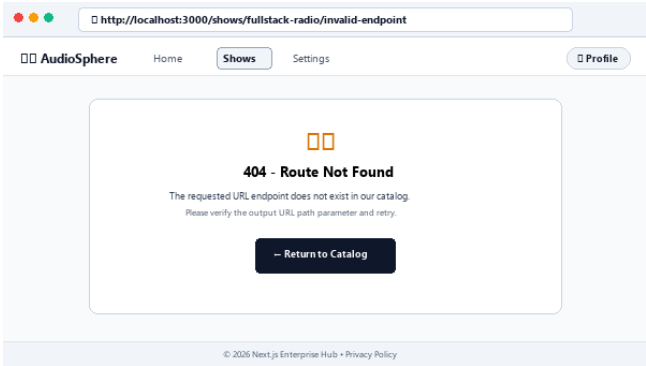
Screen 2: Detail View (/shows/fullstack-radio/episodes/react-19)



Screen 3: Shimmer Skeleton (loading.tsx)



Screen 4: Custom 404 Fallback (not-found.tsx)



Academic Tasks CSE326 (26271)

Question 11 — SupportDesk — SLA Helpdesk Portal

Scenario: Build an enterprise SLA Helpdesk Portal implementing Next.js App Router Server Actions + Form Validation. Users navigate across primary catalogs, detailed inspection endpoints, and interactive mutation actions with custom loading skeletons and 404 fallbacks.

Target Route Architecture & Expected Functionality:

Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/helpdesk/tickets?status=open	Primary application overview route implementing Server Actions + Form Validation data fetching and filtering.
/helpdesk/tickets/TICK-901	Detailed view rendering specifications, credentials, and interactive client actions.
/helpdesk/tickets/new	Dedicated action endpoint or custom error / fallback view.

Given Visual Outputs (Reproduce Exactly as Shown Below):

Screen 1: Primary View (/helpdesk/tickets?status=open)

SupportDesk

Home

Tickets

Settings

Profile

SupportDesk Overview

Next.js Architecture: Server Actions + Form Vali

Filter records...

ID / CODE	RECORD / DESCRIPTION	STATUS	ACTION
TICK-901	Production Database Latency Spik	Critical	View --
TICK-902	OAuth2 Callback Timeout Error	High	View --
TICK-903	Stripe Webhook Signature Mismatch	High	View --
TICK-904	Tailwind CSS Layout Misalignment	Low	View --

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Screen 2: Detail View (/helpdesk/tickets/TICK-901)

SupportDesk

Home

Tickets

Settings

Profile

Back to Overview /helpdesk/tickets/TICK-901

SupportDesk Inspection

Architecture: Server Actions + Form Vali

SPECIFICATIONS & ARCHITECTURE:

1. Server Actions + Form Validation

2. Layout State Continuity

3. Shimmer Loading Skeleton

4. Custom 404 Not Found Fallback

Execution Summary

Rendering: Dynamic SSR/SSG

Data Fetch: Server Fetch API

Pricing / Tag: \$49 USD

Verification: Passed (100%)

Bookmarked / Saved

Execute Action

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Screen 3: Shimmer Skeleton (loading.tsx)

SupportDesk

Home

Tickets

Settings

Profile

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Screen 4: Custom 404 Fallback (not-found.tsx)

SupportDesk

Home

Tickets

Settings

Profile

404 - Route Not Found

The requested URL endpoint does not exist in our catalog.

Please verify the output URL path parameter and retry.

Return to Catalog

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Academic Tasks CSE326 (26271)

Question 12 — ReviewForge — Testimonial Hub

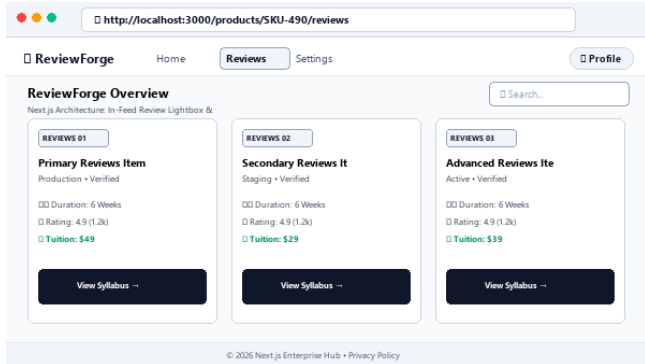
Scenario: Build an enterprise Testimonial Hub implementing Next.js App Router In-Feed Review Lightbox & Standalone Views. Users navigate across primary catalogs, detailed inspection endpoints, and interactive mutation actions with custom loading skeletons and 404 fallbacks.

Target Route Architecture & Expected Functionality:

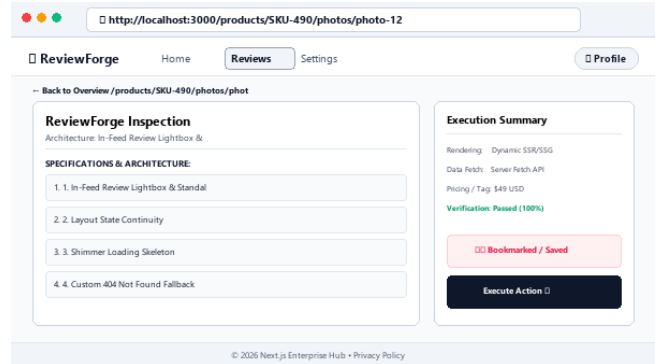
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/products/SKU-490/reviews	Primary application overview route implementing In-Feed Review Lightbox & Standalone Views data fetching and filtering.
/products/SKU-490/photos/photo-12	Detailed view rendering specifications, credentials, and interactive client actions.
/api/reviews/upvote	Dedicated action endpoint or custom error / fallback view.

Given Visual Outputs (Reproduce Exactly as Shown Below):

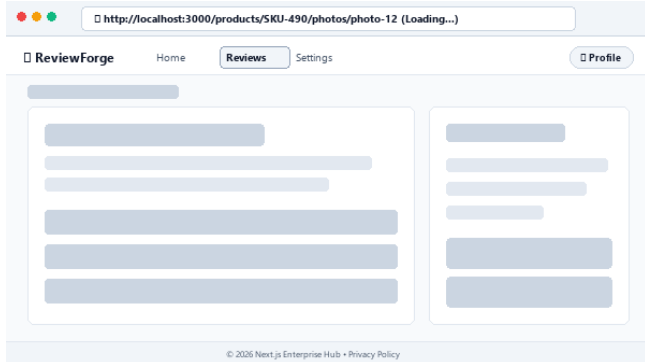
Screen 1: Primary View (/products/SKU-490/reviews)



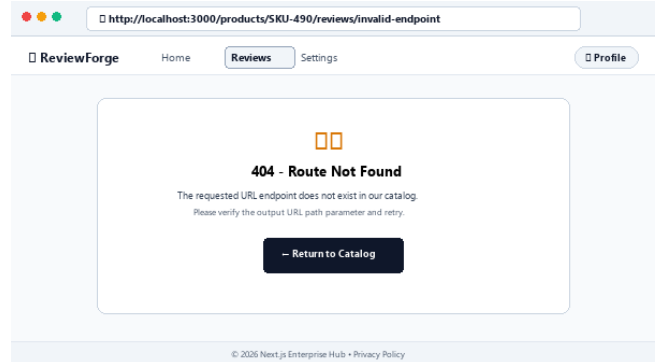
Screen 2: Detail View (/products/SKU-490/photos/photo-12)



Screen 3: Shimmer Skeleton (loading.tsx)



Screen 4: Custom 404 Fallback (not-found.tsx)



A screenshot of the Next.js Enterprise Hub web application. The browser's address bar shows the URL 'http://localhost:3000/summit/2026/schedule/invalid-endpoint'. The application's navigation bar includes a logo, the text 'EventGo', and links for 'Home', 'Summit' (which is highlighted with a blue border), 'Settings', and 'Profile'. The main content area displays a 404 error message: '404 - Route Not Found'. Below this, it states 'The requested URL endpoint does not exist in our catalog.' and provides a hint: 'Please verify the output URL path parameter and retry.' At the bottom of the error message, there is a dark blue button with the text '- Return to Catalog'.

Academic Tasks CSE326 (26271)

Question 14 — JobTrack — Candidate Hiring Wizard

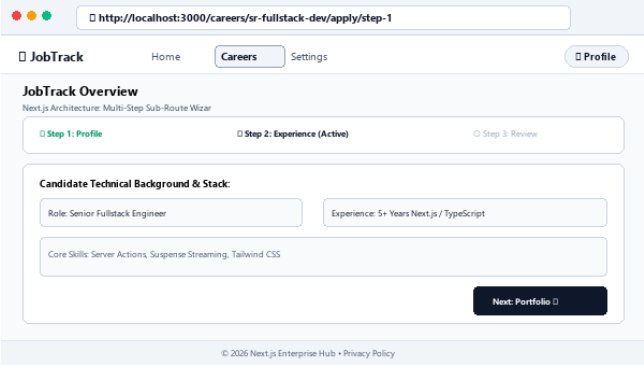
Scenario: Build an enterprise Candidate Hiring Wizard implementing Next.js App Router Multi-Step Sub-Route Wizard with Preserved State. Users navigate across primary catalogs, detailed inspection endpoints, and interactive mutation actions with custom loading skeletons and 404 fallbacks.

Target Route Architecture & Expected Functionality:

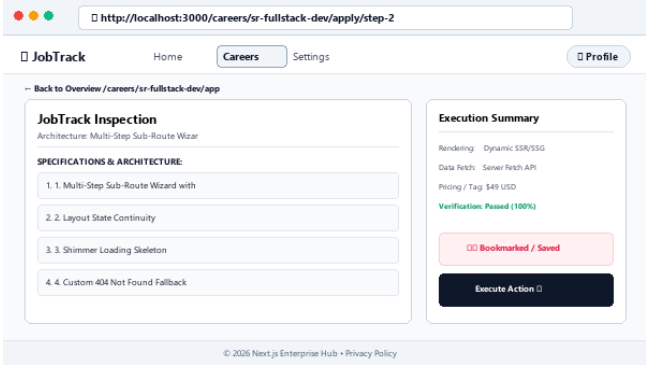
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/careers/sr-fullstack-dev/apply/step-1	Primary application overview route implementing Multi-Step Sub-Route Wizard with Preserved State data fetching and filtering.
/careers/sr-fullstack-dev/apply/step-2	Detailed view rendering specifications, credentials, and interactive client actions.
/careers/sr-fullstack-dev/apply/step-3	Dedicated action endpoint or custom error / fallback view.

Given Visual Outputs (Reproduce Exactly as Shown Below):

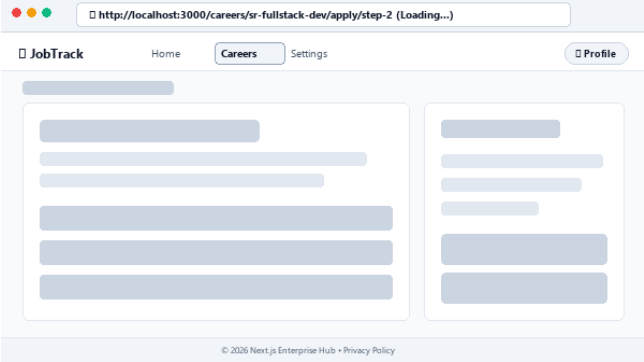
Screen 1: Primary View (/careers/sr-fullstack-dev/apply/step-1)



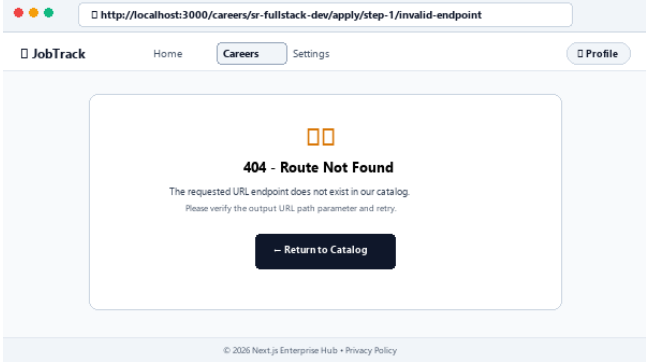
Screen 2: Detail View (/careers/sr-fullstack-dev/apply/step-2)

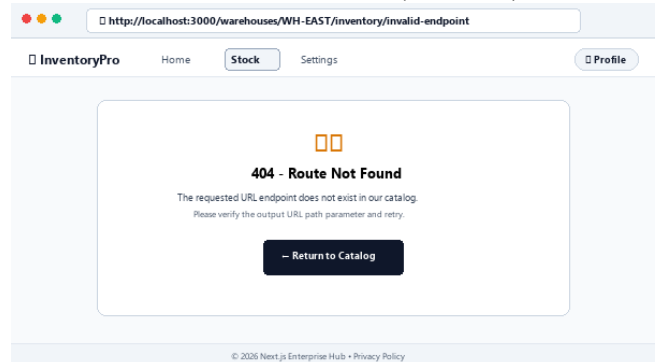


Screen 3: Shimmer Skeleton (loading.tsx)



Screen 4: Custom 404 Fallback (not-found.tsx)





Academic Tasks CSE326 (26271)

Question 16 — SaaS Metrics — Executive Analytics Console

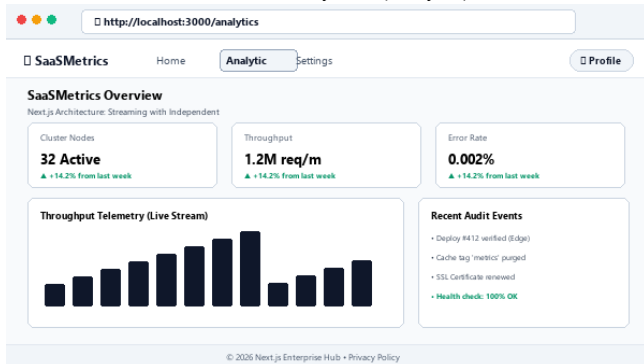
Scenario: Build an enterprise Executive Analytics Console implementing Next.js App Router Streaming with Independent Suspense Boundaries. Users navigate across primary catalogs, detailed inspection endpoints, and interactive mutation actions with custom loading skeletons and 404 fallbacks.

Target Route Architecture & Expected Functionality:

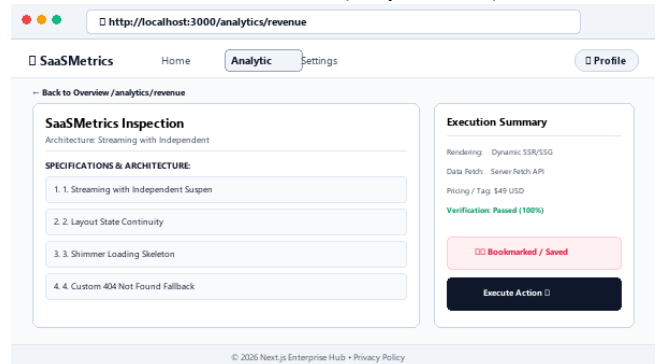
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/analytics	Primary application overview route implementing Streaming with Independent Suspense Boundaries data fetching and filtering.
/analytics/revenue	Detailed view rendering specifications, credentials, and interactive client actions.
/analytics/infrastructure	Dedicated action endpoint or custom error / fallback view.

Given Visual Outputs (Reproduce Exactly as Shown Below):

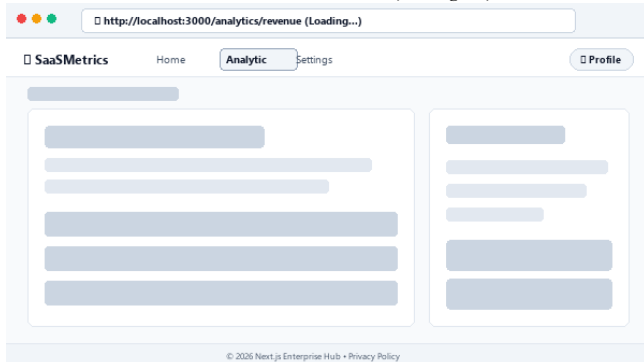
Screen 1: Primary View (/analytics)



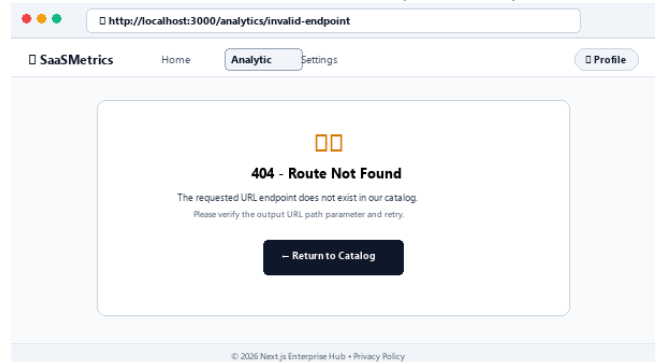
Screen 2: Detail View (/analytics/revenue)



Screen 3: Shimmer Skeleton (loading.tsx)



Screen 4: Custom 404 Fallback (not-found.tsx)



Academic Tasks CSE326 (26271)

Question 17 — CinemaScope — Film Explorer

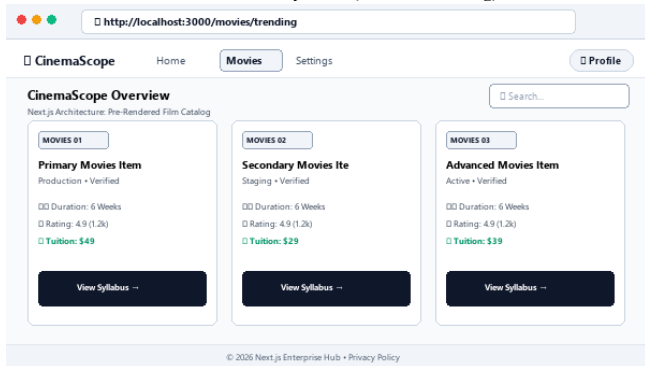
Scenario: Build an enterprise Film Explorer implementing Next.js App Router Pre-Rendered Film Catalog & Deferred Reviews. Users navigate across primary catalogs, detailed inspection endpoints, and interactive mutation actions with custom loading skeletons and 404 fallbacks.

Target Route Architecture & Expected Functionality:

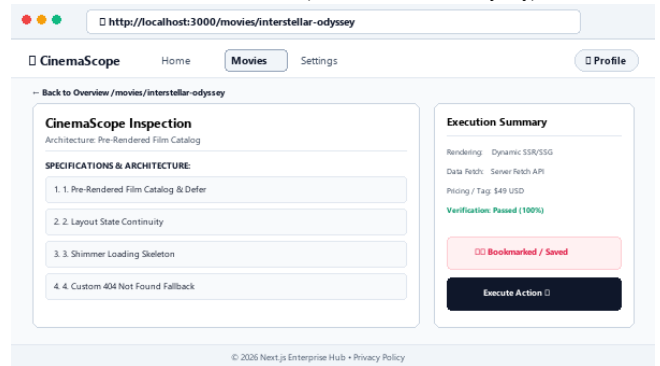
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/movies/trending	Primary application overview route implementing Pre-Rendered Film Catalog & Deferred Reviews data fetching and filtering.
/movies/interstellar-odyssey	Detailed view rendering specifications, credentials, and interactive client actions.
/movies/interstellar-odyssey/trailer	Dedicated action endpoint or custom error / fallback view.

Given Visual Outputs (Reproduce Exactly as Shown Below):

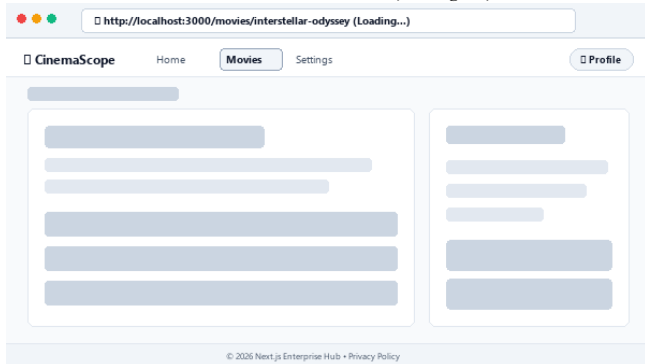
Screen 1: Primary View (/movies/trending)



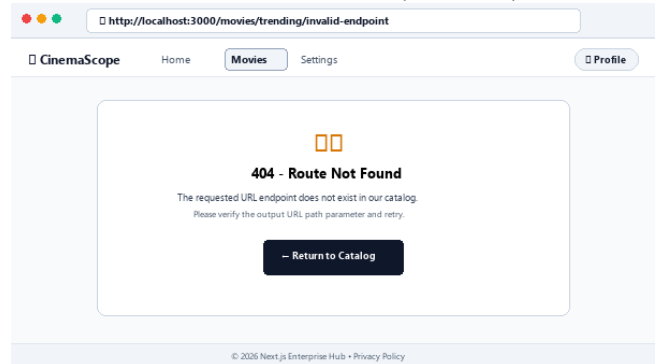
Screen 2: Detail View (/movies/interstellar-odyssey)



Screen 3: Shimmer Skeleton (loading.tsx)



Screen 4: Custom 404 Fallback (not-found.tsx)



Academic Tasks CSE326 (26271)

Question 18 — FlightRadar — Airport Live Telemetry

Scenario: Build an enterprise Airport Live Telemetry implementing Next.js App Router Granular Error Boundary with Reset Recovery. Users navigate across primary catalogs, detailed inspection endpoints, and interactive mutation actions with custom loading skeletons and 404 fallbacks.

Target Route Architecture & Expected Functionality:

Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/airports/SFO/departures	Primary application overview route implementing Granular Error Boundary with Reset Recovery data fetching and filtering.
/airports/SFO/flight/UA-882	Detailed view rendering specifications, credentials, and interactive client actions.
/airports/SFO/flight/UNKNOWN-000	Dedicated action endpoint or custom error / fallback view.

Given Visual Outputs (Reproduce Exactly as Shown Below):

Screen 1: Primary View (/airports/SFO/departures)

FlightRadar

Home

Airports

Settings

Profile

FlightRadar Overview

Next.js Architecture: Granular Error Boundary w/

ID / CODE	RECORD / DESCRIPTION	STATUS	ACTION
UA-882	Tokyo Haneda (HND) • Gate G02	Boarding	Track --
DL-402	New York (JFK) • Gate B14	On Time	Track --
BA-286	London Heathrow (LHR) • Gate A10	Delayed 20	Track --
AF-084	Paris Charles de Gaulle (CDG) •	On Time	Track --

Screen 3: Shimmer Skeleton (loading.tsx)

FlightRadar

Home

Airports

Settings

Profile

Screen 2: Detail View (/airports/SFO/flight/UA-882)

FlightRadar

Home

Airports

Settings

Profile

Back to Overview /airports/SFO/flight/UA-882

FlightRadar Inspection

Architecture: Granular Error Boundary w/

SPECIFICATIONS & ARCHITECTURE:

1. Granular Error Boundary with Reset

2. Layout State Continuity

3. Shimmer Loading Skeleton

4. Custom 404 Not Found Fallback

Execution Summary

Rendering: Dynamic SSR/SSG

Data Fetch: Server Fetch API

Pricing / Tag: \$49 USD

Verification: Passed (100%)

Bookmarked / Saved

Execute Action

Screen 4: Custom 404 Fallback (not-found.tsx)

FlightRadar

Home

Airports

Settings

Profile

404 - Route Not Found

The requested URL endpoint does not exist in our catalog.

Please verify the output URL path parameter and retry.

Return to Catalog

Academic Tasks CSE326 (26271)

Question 19 — GitPulse — Code Repository Explorer

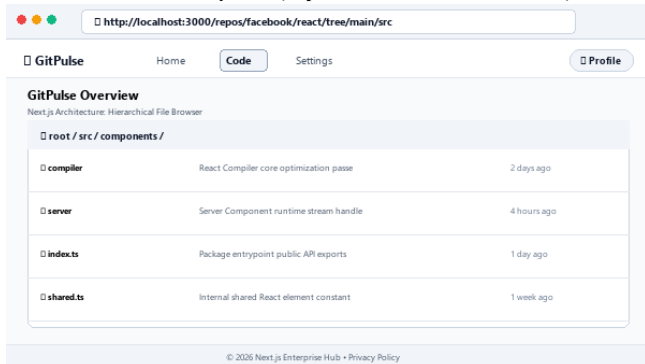
Scenario: Build an enterprise Code Repository Explorer implementing Next.js App Router Hierarchical File Browser & Shimmer Skeletons. Users navigate across primary catalogs, detailed inspection endpoints, and interactive mutation actions with custom loading skeletons and 404 fallbacks.

Target Route Architecture & Expected Functionality:

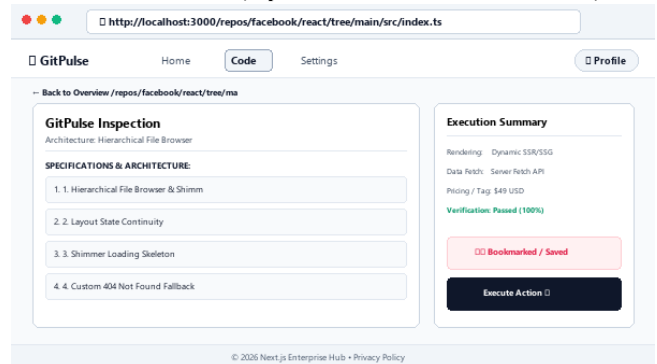
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
<code>/repos/facebook/react/tree/main/src</code>	Primary application overview route implementing Hierarchical File Browser & Shimmer Skeletons data fetching and filtering.
<code>/repos/facebook/react/tree/main/src/index.ts</code>	Detailed view rendering specifications, credentials, and interactive client actions.
<code>/repos/facebook/react/commits/main</code>	Dedicated action endpoint or custom error / fallback view.

Given Visual Outputs (Reproduce Exactly as Shown Below):

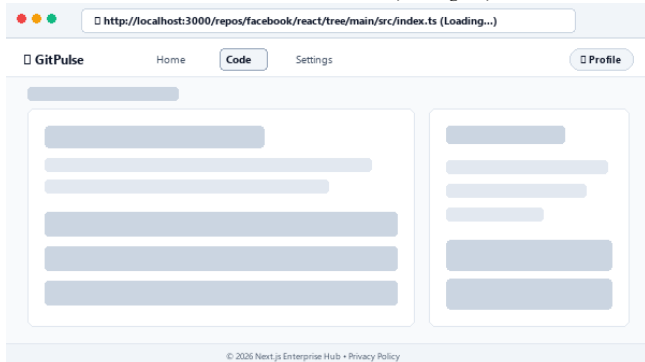
Screen 1: Primary View (`/repos/facebook/react/tree/main/src`)



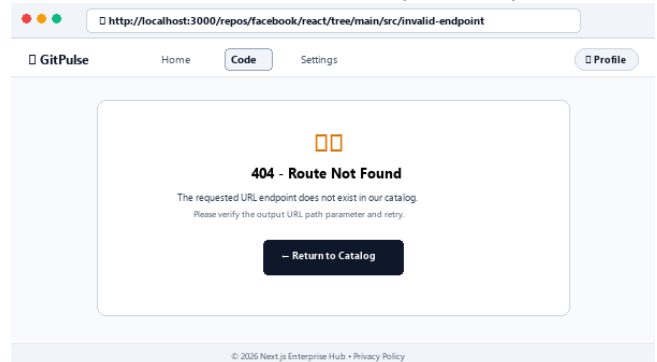
Screen 2: Detail View (`/repos/facebook/react/tree/main/src/index.ts`)



Screen 3: Shimmer Skeleton (`loading.tsx`)



Screen 4: Custom 404 Fallback (`not-found.tsx`)



Academic Tasks CSE326 (26271)

Question 20 — WeatherSphere — Global Meteorological Radar

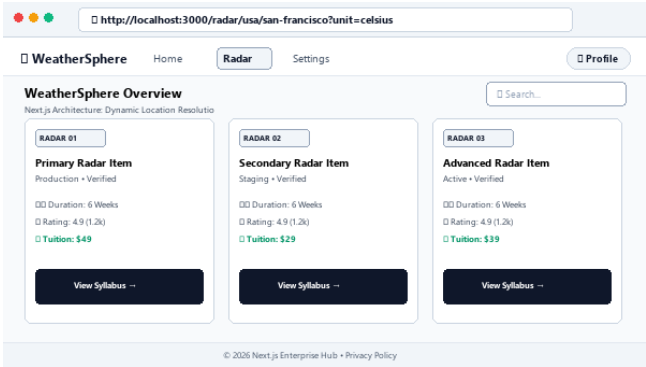
Scenario: Build an enterprise Global Meteorological Radar implementing Next.js App Router Dynamic Location Resolution & Query State. Users navigate across primary catalogs, detailed inspection endpoints, and interactive mutation actions with custom loading skeletons and 404 fallbacks.

Target Route Architecture & Expected Functionality:

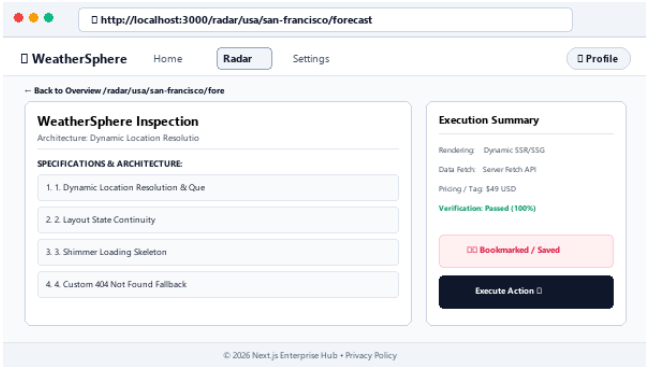
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/radar/usa/san-francisco?unit=celsius	Primary application overview route implementing Dynamic Location Resolution & Query State data fetching and filtering.
/radar/usa/san-francisco/forecast	Detailed view rendering specifications, credentials, and interactive client actions.
/radar/usa/invalid-coordinates	Dedicated action endpoint or custom error / fallback view.

Given Visual Outputs (Reproduce Exactly as Shown Below):

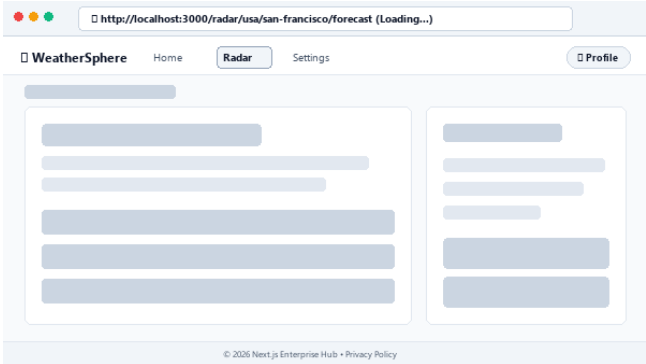
Screen 1: Primary View (/radar/usa/san-francisco?unit=celsius)



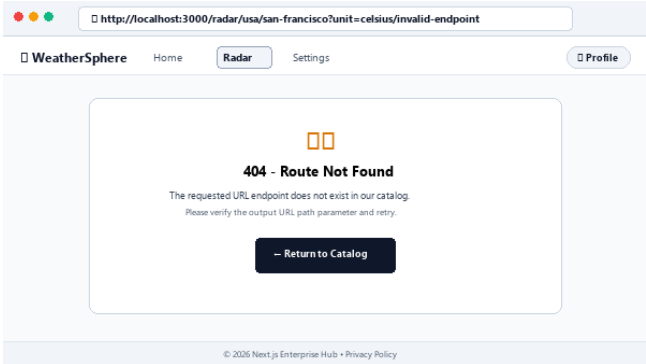
Screen 2: Detail View (/radar/usa/san-francisco/forecast)



Screen 3: Shimmer Skeleton (loading.tsx)



Screen 4: Custom 404 Fallback (not-found.tsx)



Academic Tasks CSE326 (26271)

Question 21 — InstaSnap — Photography Gallery

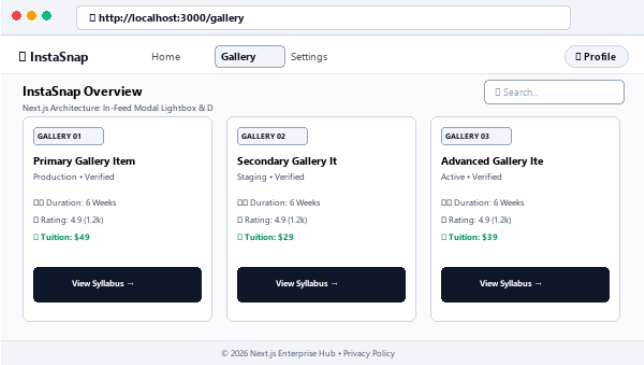
Scenario: Build an enterprise Photography Gallery implementing Next.js App Router In-Feed Modal Lightbox & Direct Full Views. Users navigate across primary catalogs, detailed inspection endpoints, and interactive mutation actions with custom loading skeletons and 404 fallbacks.

Target Route Architecture & Expected Functionality:

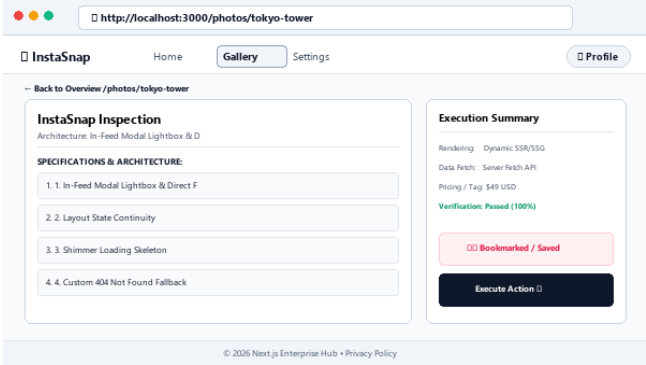
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/gallery	Primary application overview route implementing In-Feed Modal Lightbox & Direct Full Views data fetching and filtering.
/photos/tokyo-tower	Detailed view rendering specifications, credentials, and interactive client actions.
/gallery/photo-preview/tokyo-tower	Dedicated action endpoint or custom error / fallback view.

Given Visual Outputs (Reproduce Exactly as Shown Below):

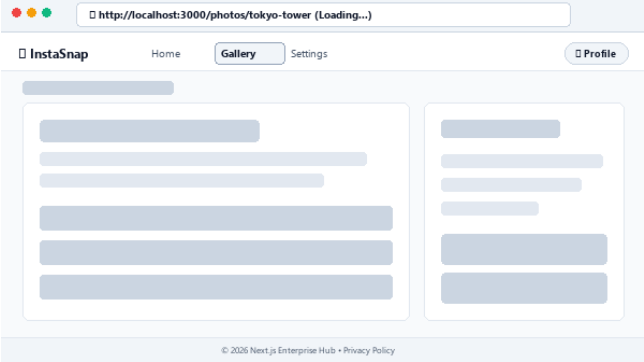
Screen 1: Primary View (/gallery)



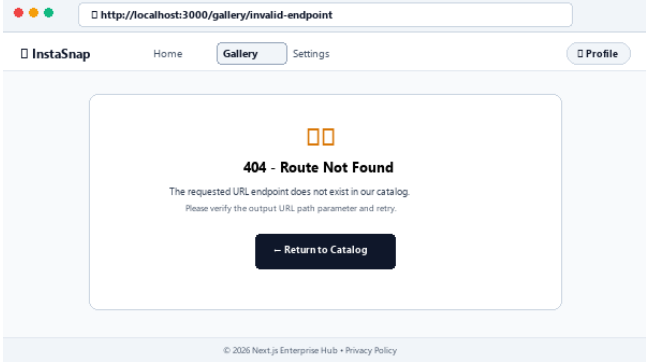
Screen 2: Detail View (/photos/tokyo-tower)



Screen 3: Shimmer Skeleton (loading.tsx)



Screen 4: Custom 404 Fallback (not-found.tsx)



Academic Tasks CSE326 (26271)

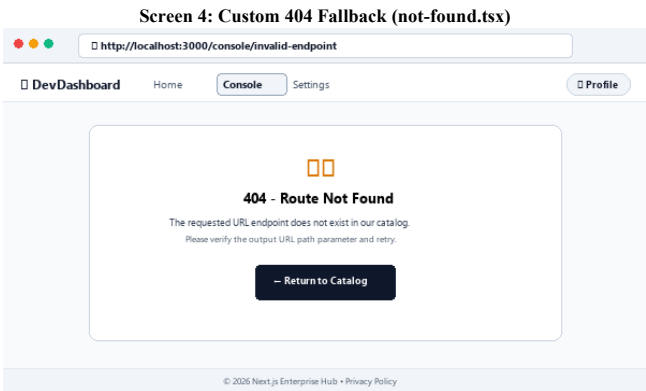
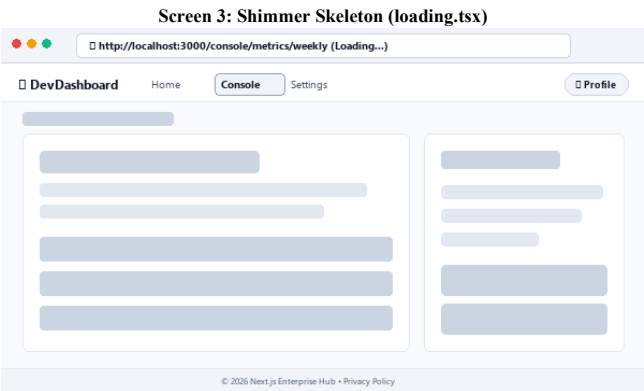
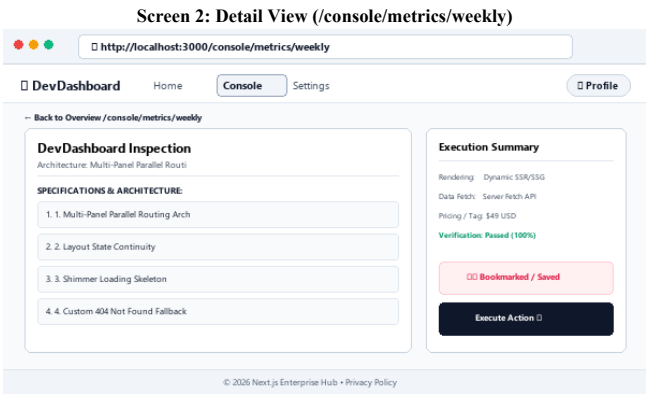
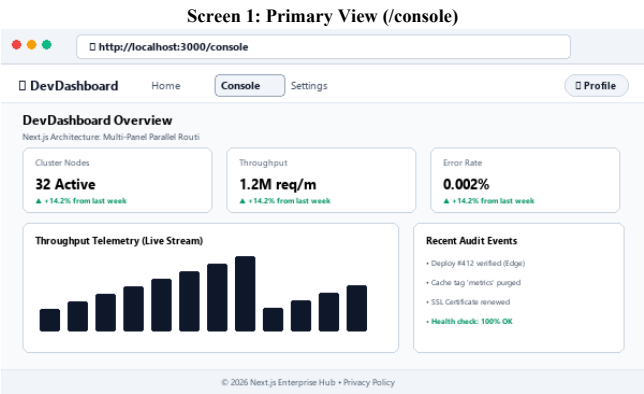
Question 22 — DevDashboard — Multi-Slot Parallel Console

Scenario: Build an enterprise Multi-Slot Parallel Console implementing Next.js App Router Multi-Panel Parallel Routing Architecture. Users navigate across primary catalogs, detailed inspection endpoints, and interactive mutation actions with custom loading skeletons and 404 fallbacks.

Target Route Architecture & Expected Functionality:

Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/console	Primary application overview route implementing Multi-Panel Parallel Routing Architecture data fetching and filtering.
/console/metrics/weekly	Detailed view rendering specifications, credentials, and interactive client actions.
/console/audit-logs/security	Dedicated action endpoint or custom error / fallback view.

Given Visual Outputs (Reproduce Exactly as Shown Below):



Academic Tasks CSE326 (26271)

Question 23 — CloudDrive — Storage Browser

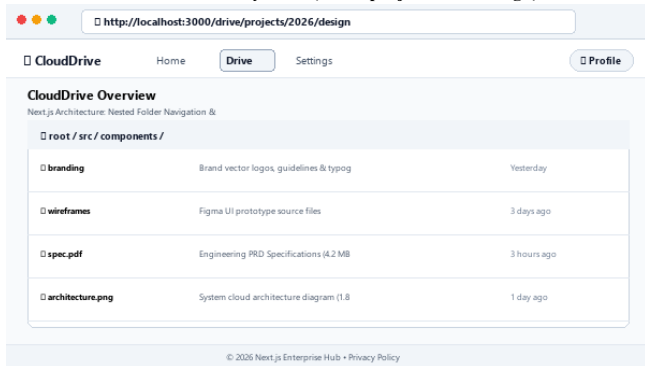
Scenario: Build an enterprise Storage Browser implementing Next.js App Router Nested Folder Navigation & Slide-Out Drawer. Users navigate across primary catalogs, detailed inspection endpoints, and interactive mutation actions with custom loading skeletons and 404 fallbacks.

Target Route Architecture & Expected Functionality:

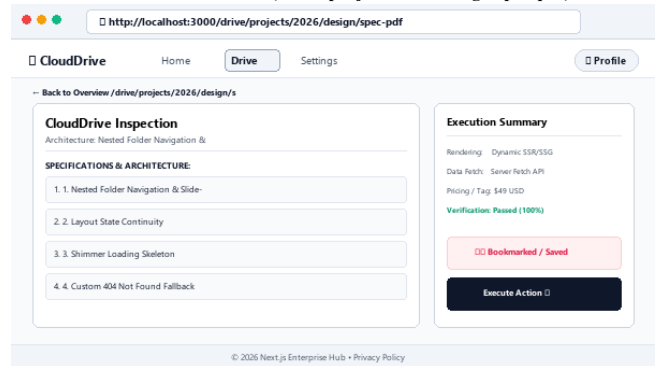
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/drive/projects/2026/design	Primary application overview route implementing Nested Folder Navigation & Slide-Out Drawer data fetching and filtering.
/drive/projects/2026/design/spec-pdf	Detailed view rendering specifications, credentials, and interactive client actions.
/drive/trash	Dedicated action endpoint or custom error / fallback view.

Given Visual Outputs (Reproduce Exactly as Shown Below):

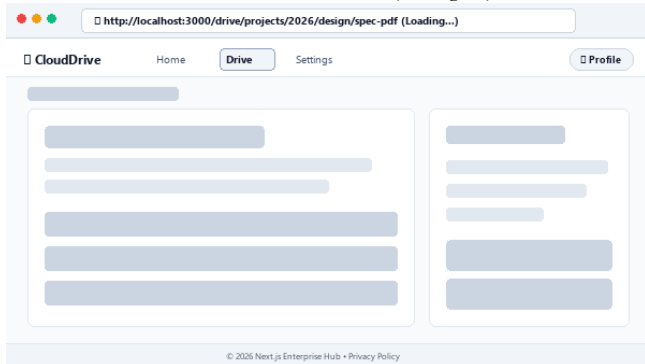
Screen 1: Primary View (/drive/projects/2026/design)



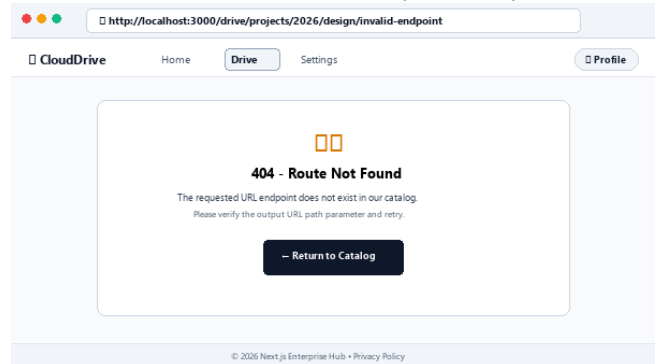
Screen 2: Detail View (/drive/projects/2026/design/spec-pdf)



Screen 3: Shimmer Skeleton (loading.tsx)



Screen 4: Custom 404 Fallback (not-found.tsx)



Academic Tasks CSE326 (26271)

Question 24 — KeyVault — API Key Credential Manager

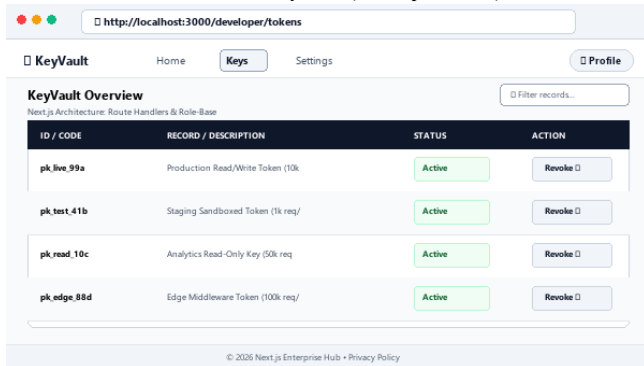
Scenario: Build an enterprise API Key Credential Manager implementing Next.js App Router Route Handlers & Role-Based Access Protection. Users navigate across primary catalogs, detailed inspection endpoints, and interactive mutation actions with custom loading skeletons and 404 fallbacks.

Target Route Architecture & Expected Functionality:

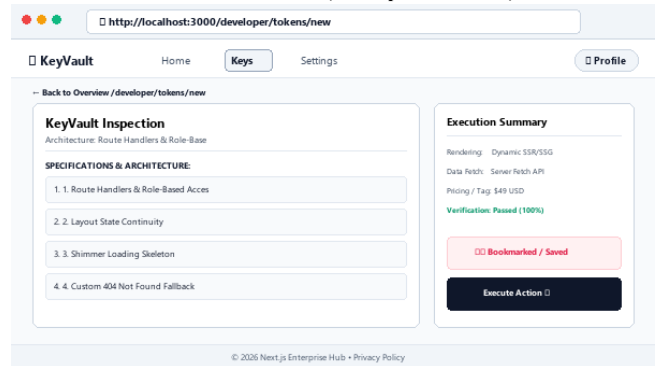
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/developer/tokens	Primary application overview route implementing Route Handlers & Role-Based Access Protection data fetching and filtering.
/developer/tokens/new	Detailed view rendering specifications, credentials, and interactive client actions.
/api/tokens/revoke	Dedicated action endpoint or custom error / fallback view.

Given Visual Outputs (Reproduce Exactly as Shown Below):

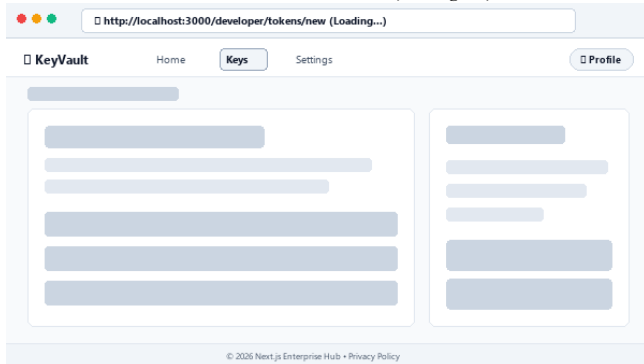
Screen 1: Primary View (/developer/tokens)



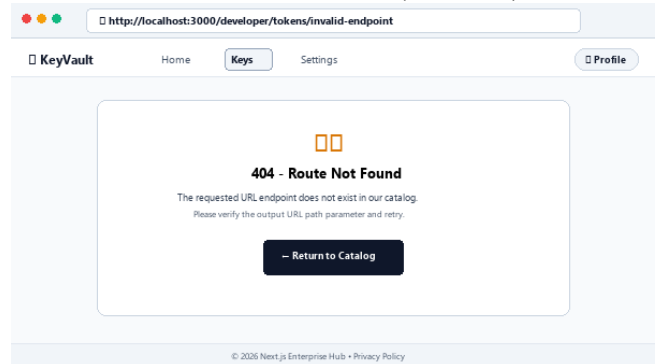
Screen 2: Detail View (/developer/tokens/new)



Screen 3: Shimmer Skeleton (loading.tsx)



Screen 4: Custom 404 Fallback (not-found.tsx)



Academic Tasks CSE326 (26271)

Question 25 — CartDrawer — E-Commerce Storefront

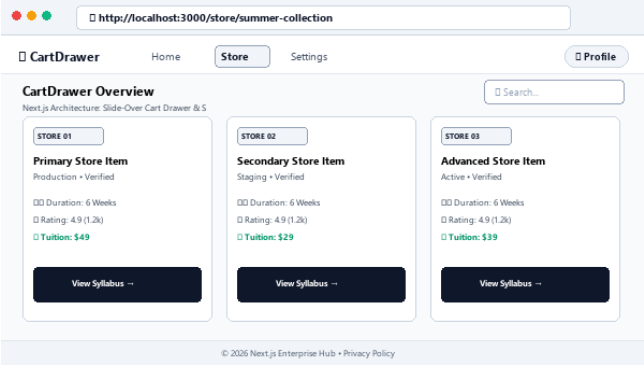
Scenario: Build an enterprise E-Commerce Storefront implementing Next.js App Router Slide-Over Cart Drawer & Standalone Checkout. Users navigate across primary catalogs, detailed inspection endpoints, and interactive mutation actions with custom loading skeletons and 404 fallbacks.

Target Route Architecture & Expected Functionality:

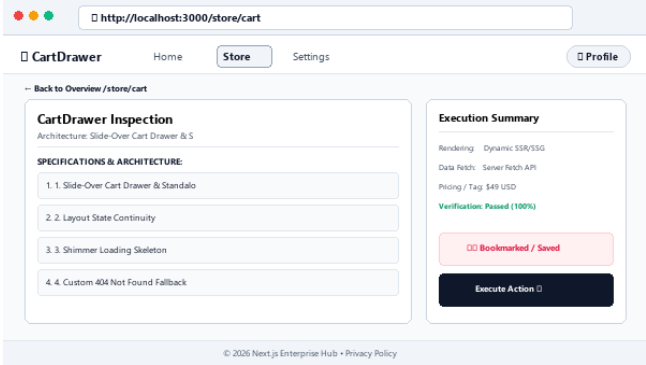
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/store/summer-collection	Primary application overview route implementing Slide-Over Cart Drawer & Standalone Checkout data fetching and filtering.
/store/cart	Detailed view rendering specifications, credentials, and interactive client actions.
/checkout	Dedicated action endpoint or custom error / fallback view.

Given Visual Outputs (Reproduce Exactly as Shown Below):

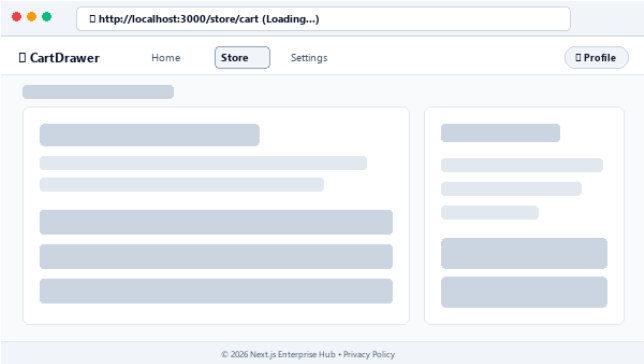
Screen 1: Primary View (/store/summer-collection)



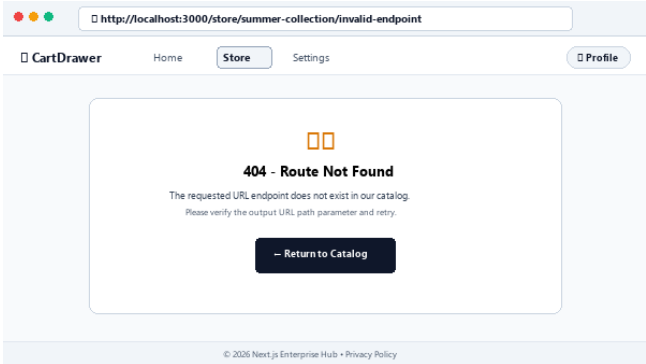
Screen 2: Detail View (/store/cart)



Screen 3: Shimmer Skeleton (loading.tsx)



Screen 4: Custom 404 Fallback (not-found.tsx)



Academic Tasks CSE326 (26271)

Question 26 — TechNews — Breaking News Engine

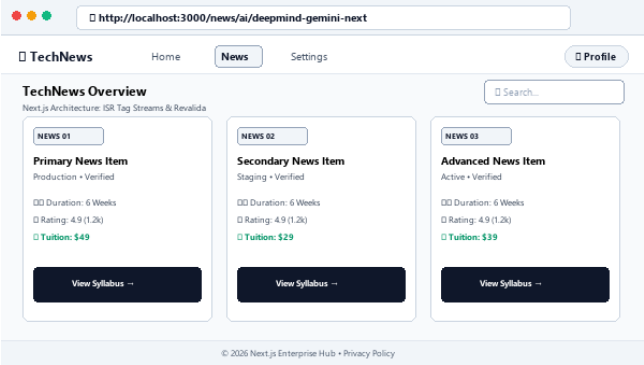
Scenario: Build an enterprise Breaking News Engine implementing Next.js App Router ISR Tag Streams & Revalidated News Ticker. Users navigate across primary catalogs, detailed inspection endpoints, and interactive mutation actions with custom loading skeletons and 404 fallbacks.

Target Route Architecture & Expected Functionality:

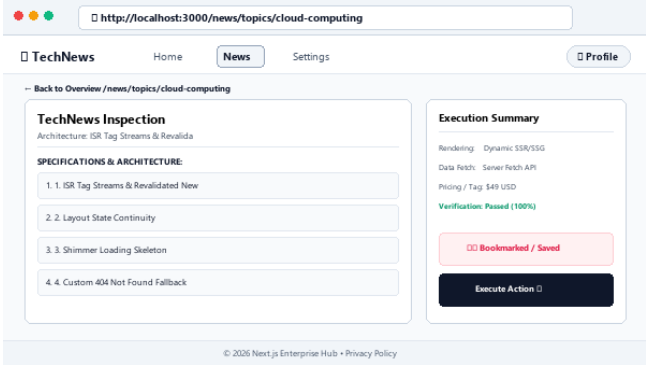
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/news/ai/deepmind-gemini-next	Primary application overview route implementing ISR Tag Streams & Revalidated News Ticker data fetching and filtering.
/news/topics/cloud-computing	Detailed view rendering specifications, credentials, and interactive client actions.
/api/news/breaking-ticker	Dedicated action endpoint or custom error / fallback view.

Given Visual Outputs (Reproduce Exactly as Shown Below):

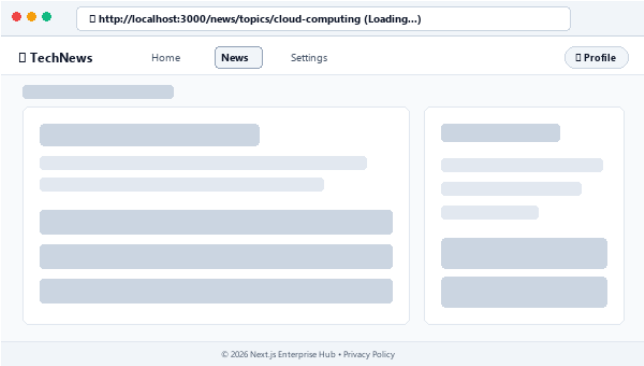
Screen 1: Primary View (/news/ai/deepmind-gemini-next)



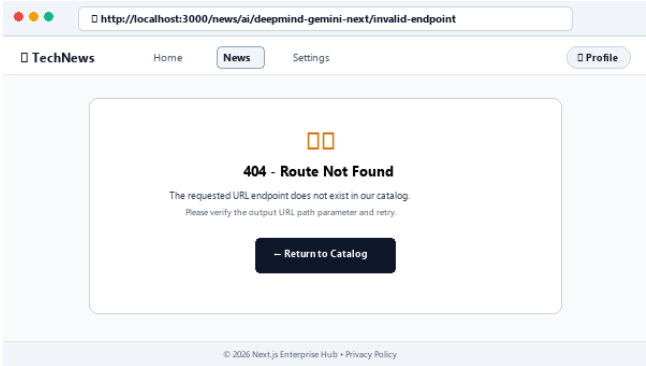
Screen 2: Detail View (/news/topics/cloud-computing)

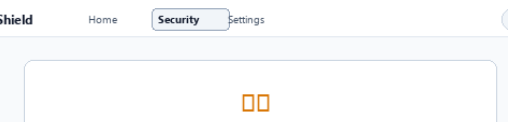


Screen 3: Shimmer Skeleton (loading.tsx)



Screen 4: Custom 404 Fallback (not-found.tsx)





The screenshot shows a web browser window with the address bar displaying `http://localhost:3000/admin/security-audit/invalid-endpoint`. The page has a navigation bar with links for **AuthShield**, **Home**, **Security** (which is highlighted), and **Settings**. On the right side of the navigation bar, there is a **Profile** link. The main content area is a light gray box containing a 404 error message. At the top of the message box are two orange squares. Below them, the text reads **404 - Route Not Found**. Underneath, it says: "The requested URL endpoint does not exist in our catalog. Please verify the output URL path parameter and retry." At the bottom of the message box is a dark blue button with the text **Return to Catalog**.

Academic Tasks CSE326 (26271)

Question 28 — GlobalHub — Multi-Lingual Storefront

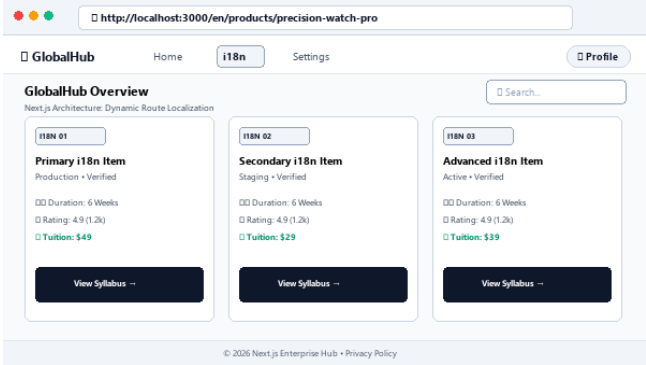
Scenario: Build an enterprise Multi-Lingual Storefront implementing Next.js App Router Dynamic Route Localization & Currency Adapters. Users navigate across primary catalogs, detailed inspection endpoints, and interactive mutation actions with custom loading skeletons and 404 fallbacks.

Target Route Architecture & Expected Functionality:

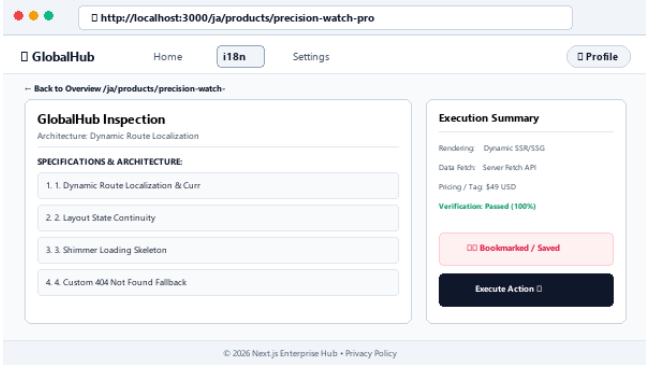
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/en/products/precision-watch-pro	Primary application overview route implementing Dynamic Route Localization & Currency Adapters data fetching and filtering.
/ja/products/precision-watch-pro	Detailed view rendering specifications, credentials, and interactive client actions.
/en/checkout	Dedicated action endpoint or custom error / fallback view.

Given Visual Outputs (Reproduce Exactly as Shown Below):

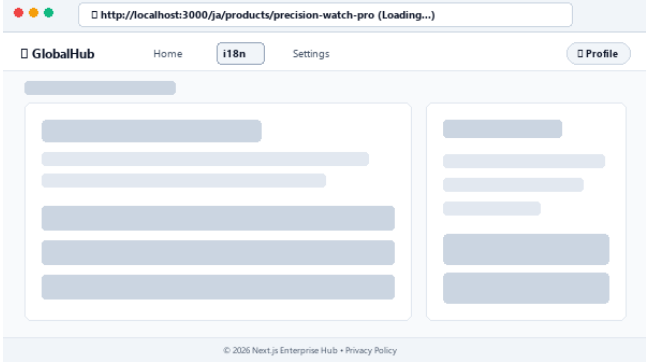
Screen 1: Primary View (/en/products/precision-watch-pro)



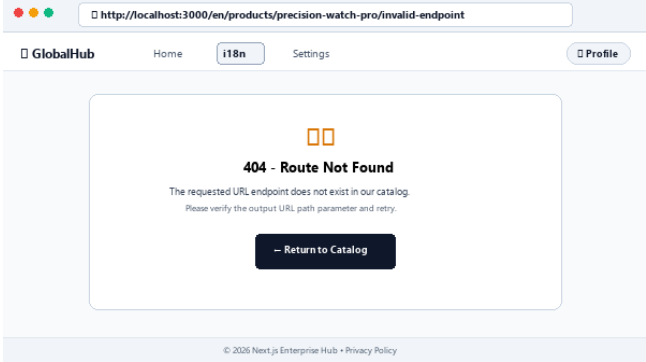
Screen 2: Detail View (/ja/products/precision-watch-pro)



Screen 3: Shimmer Skeleton (loading.tsx)



Screen 4: Custom 404 Fallback (not-found.tsx)



Academic Tasks CSE326 (26271)

Question 29 — SEOMaster — Technical Publishing Platform

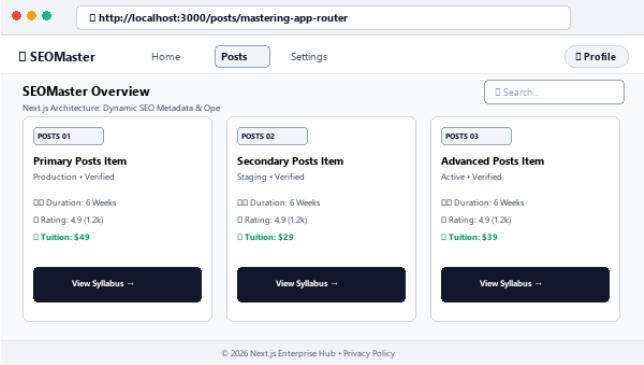
Scenario: Build an enterprise Technical Publishing Platform implementing Next.js App Router Dynamic SEO Metadata & OpenGraph Banner Route. Users navigate across primary catalogs, detailed inspection endpoints, and interactive mutation actions with custom loading skeletons and 404 fallbacks.

Target Route Architecture & Expected Functionality:

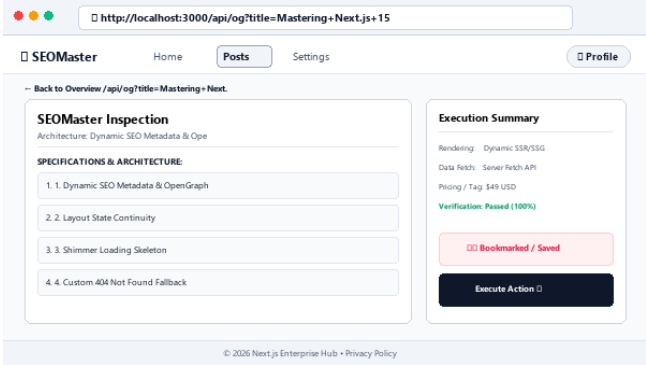
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/posts/mastering-app-router	Primary application overview route implementing Dynamic SEO Metadata & OpenGraph Banner Route data fetching and filtering.
/api/og?title=Mastering+Next.js+15	Detailed view rendering specifications, credentials, and interactive client actions.
/posts/mastering-app-router/share-preview	Dedicated action endpoint or custom error / fallback view.

Given Visual Outputs (Reproduce Exactly as Shown Below):

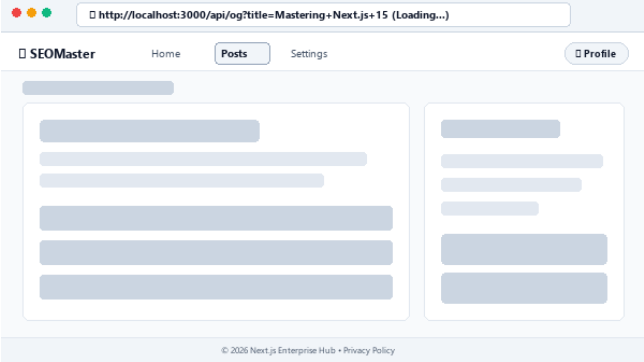
Screen 1: Primary View (/posts/mastering-app-router)



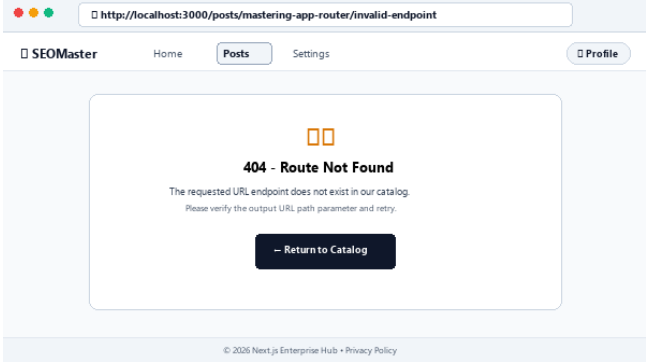
Screen 2: Detail View (/api/og?title=Mastering+Next.js+15)



Screen 3: Shimmer Skeleton (loading.tsx)



Screen 4: Custom 404 Fallback (not-found.tsx)



Academic Tasks CSE326 (26271)

Question 30 — TenantCloud — Multi-Tenant SaaS Workspace

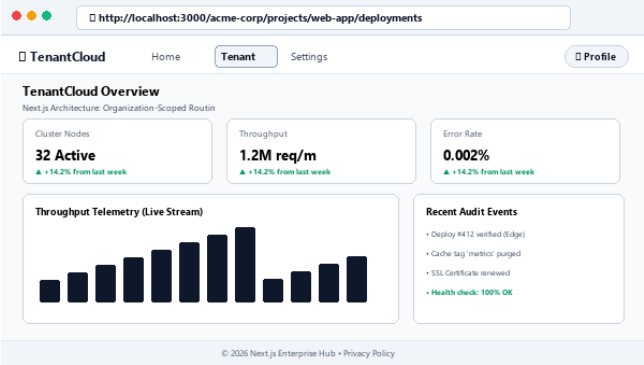
Scenario: Build an enterprise Multi-Tenant SaaS Workspace implementing Next.js App Router Organization-Scoped Routing & Theming Isolation. Users navigate across primary catalogs, detailed inspection endpoints, and interactive mutation actions with custom loading skeletons and 404 fallbacks.

Target Route Architecture & Expected Functionality:

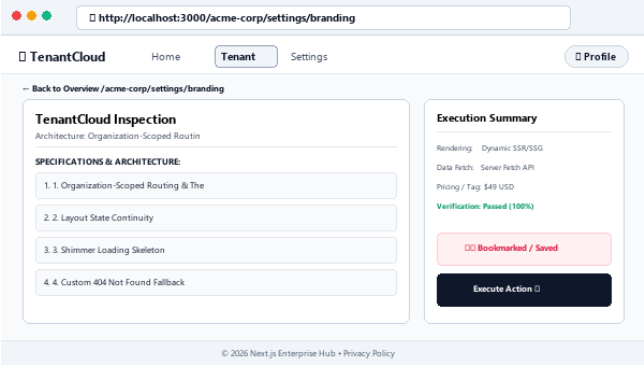
Target Route URL / Endpoint	Expected User Flow & Functional Behavior
/acme-corp/projects/web-app/deployments	Primary application overview route implementing Organization-Scoped Routing & Theming Isolation data fetching and filtering.
/acme-corp/settings/branding	Detailed view rendering specifications, credentials, and interactive client actions.
/acme-corp/projects/web-app/rollback	Dedicated action endpoint or custom error / fallback view.

Given Visual Outputs (Reproduce Exactly as Shown Below):

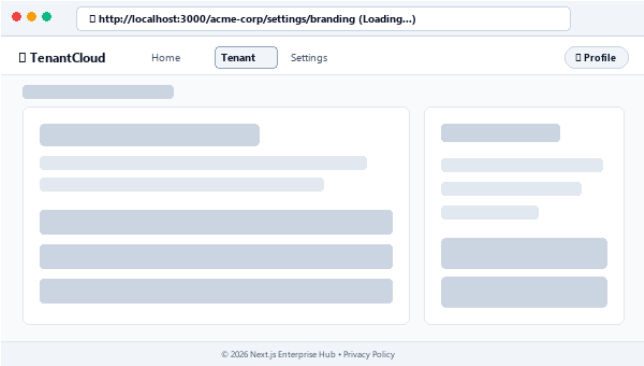
Screen 1: Primary View (/acme-corp/projects/web-app/deployments)



Screen 2: Detail View (/acme-corp/settings/branding)



Screen 3: Shimmer Skeleton (loading.tsx)



Screen 4: Custom 404 Fallback (not-found.tsx)

